DATA NARRATIVES · GROUP PROJECT

Cormorants Spirits: Sales Mix Analysis

A Claude-Integrated Data Storytelling Assignment

Design Philosophy

This assignment is built around a single provocation from a colleague in our department: if AI can produce the artifact, the artifact should not be the assessment. Your deliverables — the report, the slides, the storyboard — are evidence of preparation. They are not the measure of learning. The measure of learning is whether you can explain, defend, and revise your thinking in real time, under pressure, when the ground shifts.

To reflect that standard, this assignment has two evaluation layers. The first is preparation: your Claude workflow, your written report, and your presentation slides. The second is performance: a live defense conducted immediately after your presentation, where the instructor introduces a new constraint and your team must respond without a script. Both layers are graded. Neither can substitute for the other.

THE THREE LEVELS OF DATA ANALYSIS

LEVEL 1	**What does the data say?**	— Identify patterns, outliers, trends. Claude can do this. It is table stakes.
LEVEL 2	**So what?**	— Interpret the cause and context of the patterns. This requires judgment.
LEVEL 3	**Now what?**	— Make a decision and defend the trade-offs. This is where learning lives.

Level 1 earns no points on its own. Your Claude process must reach Level 2. Your Presentation Defense will be evaluated at Level 3.

Overview

You are an analytics consulting team hired by Cormorants Spirits, a producer and seller of adult spirits looking to improve their sales performance over the next two years. Your team will analyze one full year of transaction-level sales data, identify which products deserve greater emphasis and which should be reconsidered, and communicate your findings through a written executive report and a polished PowerPoint presentation — which you will then defend live in class.

Learning Objectives

1. **Interrogate data before narrativizing it.** Use Claude to ask rigorous questions and stress-test hypotheses before forming conclusions.
2. **Move through all three levels of analysis.** Reach Level 3 — a defensible decision with acknowledged trade-offs — not just a data summary.
3. **Build a defensible, data-grounded story.** Apply the Minto Principle (SCQA) to structure a clear narrative that traces directly back to the evidence.
4. **Communicate with professional clarity.** Deliver a polished one-page executive report and a Kawasaki-format presentation.

5. **Defend your reasoning in real time.** Explain, revise, and justify your thinking live — without a script — when new information is introduced.

The Scenario

Cormorants is a producer and seller of adult spirits. The company carries a broad portfolio ranging from high-volume commodity items to premium specialty spirits, and is looking for strategic guidance on how to optimize its sales mix going forward. Leadership wants to know: which products should Cormorants lean into over the next two years, and which should they consider exiting? Your team has one year of transaction-level data and the tools to make sense of it. What you present must be clear, defensible, and actionable — and you must be able to defend it when the ground shifts.

The Data

You will receive one Excel file: Cormorants Data.xlsx. It contains approximately 14,200 rows of sales transactions from a single calendar year. The key fields are:

- **Customer Name:** 42 anonymized customer accounts
- **Sales Date:** Transaction date — useful for identifying trends across the year
- **Item Class:** 18 product lines — your primary analytical dimension
- **Item Description:** Specific SKU-level product name
- **Cases:** Number of cases sold per transaction
- **Proof:** Alcohol proof of the product
- **Size:** Bottle size (50 mL through 1.75 L)
- **Bot/cs:** Bottles per case
- **Gross Rev:** Gross revenue in dollars for that transaction row

Upload the raw Excel file directly to your Claude conversation. Do not pre-aggregate or filter it first.

Required Process: Using Claude

The following four-phase workflow is required. Each numbered prompt must appear in your submitted Claude conversation log. You may expand on these prompts — in fact, you are expected to. But skipping phases or using fewer prompts than listed will cost you points on the Process grade.

Rule: Do not begin writing your Storyboard, Report, or Slides until Phase 3. Phases 1 and 2 earn the right to a conclusion.

PHASE 1 Data Discovery

Before forming any hypothesis, get oriented. These prompts surface patterns — not conclusions.

PROMPT 1 Orient Me
"I'm uploading a sales dataset. Before I ask any analytical questions, describe what you observe: the structure, time period, number of records, key dimensions, and any surface-level patterns. Do not make strategic recommendations yet."

PROMPT 2 Revenue by Product Line
"What are the top 5 and bottom 5 product lines (Item Class) by total gross revenue? Give me exact dollar amounts and their share of total revenue — not just rankings."

PROMPT 3 **Volume vs. Revenue Mismatch**
"Compare total cases sold to total gross revenue for each product line. Which lines have high volume but low revenue? Which have low volume but high revenue? Flag any where the ratio seems unusual."

PROMPT 4 **Trend Over Time**
"For the top 3 and bottom 3 product lines by revenue, show how monthly sales trended across the year. Are any growing, declining, or seasonal?"

PROMPT 5 **Anomalies and Outliers**
"Are there any product lines, customers, or transactions that look anomalous — unusual spikes, gaps, or data points that don't fit the pattern? Flag these before I draw any conclusions."

PHASE 2 Challenging the Story

You likely have early instincts. This phase stress-tests them before you commit to a narrative.

PROMPT 6 **Play Devil's Advocate**
"Based on what we've found, I'm leaning toward recommending that Cormorants focus on [describe your current hypothesis]. Push back on this. What would be lost if they followed this strategy? What risks or counterarguments does the data suggest?"

PROMPT 7 **What Can't This Data Tell Us?**
"What important questions about Cormorants' business can this dataset NOT answer? What limitations should I acknowledge before drawing conclusions?"

PROMPT 8 **The Strongest Case Against My Narrative**
"Give me the most compelling argument someone could make AGAINST the recommendations I seem to be building. Use the data. I want to understand what I might be getting wrong."

PHASE 3 Building the Narrative (Minto / SCQA)

Now you have earned the right to structure your story. Use Claude to help build the SCQA — but the judgment about what the story IS must be yours.

PROMPT 9 **Build the SCQA**
"Help me draft a Minto Principle SCQA structure. The Situation is: [one sentence]. Based on the data, what is the most compelling Complication, Question, and Answer? Give me two or three options for each element so I can choose."

PROMPT 10 **Support My Recommendation**
"My main recommendation is: [state it in one sentence]. What are the three strongest supporting arguments the data provides? State each as a single declarative sentence an executive would find credible."

PROMPT 11 **Identify Gaps in My Logic**
"Review this narrative: [paste your SCQA and three supporting arguments]. Does every claim trace back to the data? Flag anything that feels like a stretch, is not directly supported, or requires an assumption the data doesn't justify."

PHASE 4 Refining Your Work

Use Claude as a final editor and sounding board before you submit.

PROMPT 12 **Stress-Test the Report**
"Here is my draft executive report: [paste text]. Act as a skeptical executive reader. Where is the logic weakest? Where am I asserting something the data doesn't clearly support? Where could I be more specific?"

PROMPT 13 **Slide Alignment Check**
"Here are the titles and key takeaways for my 10 slides: [list them]. Does each carry a clear, distinct message? Rate the overall narrative arc from 1-10 and explain your rating."

PROMPT 14 **Prepare for the Shock**
"What are the three most likely 'what if' scenarios a skeptical executive could throw at us during our presentation that would force us to revise our recommendation? For each, help me think through how our core argument holds — or doesn't."

Deliverables

Your team will submit five items before class, then perform one live event during class. All are required.

1. Claude Chat Log Export

Export your full Claude conversation as a PDF. This is the evidence of your analytical process. Graders will look for: all 15 required prompts, meaningful follow-up questions beyond the scaffolded prompts, and evidence that you engaged critically with Claude's outputs rather than accepting them passively. Conversations that only show the scaffolded prompts with no follow-up will receive partial credit.

To export: In Claude, click the three-dot menu → Download → PDF.

2. Storyboard — Minto Principle (SCQA Format)

One page maximum. Use the SCQA structure: Situation, Complication, Question, Answer. Below the SCQA, include your key line and two to three supporting arguments — each stated as a single declarative sentence. This should reflect your decisions, not Claude's first draft.

3. Written Executive Report

One page maximum. Data visualizations, if included, do not count toward the page limit. Open with a clear governing thought (your answer), followed by three supporting arguments grounded in specific data. Avoid vague language. An executive reading this should know exactly what Cormorants should do and why. This is a Level 2 and Level 3 document — do not waste space on Level 1 summaries Claude already produced.

4. PowerPoint Presentation — Kawasaki Format

Maximum 10 slides. Cover counts as slide 1; Q&A counts as the last slide. Follow Kawasaki format: one idea per slide, visual over text, no full sentences in the body. Use the Notes section to write exactly how you would talk to each slide — these notes are graded. Remember: the slides are preparation. The defense is the performance.

5. Claude Reflection

250–350 words. Answer all four questions:

- At what point in the process did Claude's output most surprise you — and what did you do about it?
- Describe one specific instance where you disagreed with or pushed back on Claude. What was the outcome?

- Where did Claude add the most genuine value in your analysis? Where did it fall short?
- What would you do differently in your Claude workflow if you did this assignment again?

Reflections that only describe what you did — rather than critically evaluating it — will receive partial credit.

6. Presentation Defense — The Scenario Shock [IN CLASS]

Immediately after your team presents the deck, the instructor will introduce a Scenario Shock: a new constraint that was not in the original data and could affect your recommendation. Your team has two minutes to confer, then five minutes to defend your revised position aloud. This portion is not scripted, not submitted in advance, and cannot be prepared by AI.

EXAMPLE SHOCK SCENARIOS

- Cormorants has just lost its single largest customer, who accounted for 25% of revenue in your recommended product line. Does your recommendation hold?
- A regulatory change will require all spirits above 90 proof to carry additional labeling costs, raising their per-case cost by 15%. How does this affect your analysis?
- A competitor has offered to acquire Cormorants' lowest-performing product line, including equipment and brand rights, for a premium price. Should they take the deal?
- Leadership tells you the sales team's commission structure heavily favors the product lines you are recommending they exit. How do you account for that in your plan?

Your team will not know which scenario is coming until the moment it is revealed. You will be assessed on the quality of your reasoning, not whether your recommendation changes — changing your recommendation in light of new evidence is a sign of good thinking, not a weakness.

What the Defense Is Graded On

- **Logic & Coherence:** Does the argument follow a clear, logical progression given the new constraint?
- **Evidence Application:** Does the team reference specific data from the analysis to support their revised position?
- **Framework Usage:** Are course concepts (Minto, Dykes, Tufte) applied correctly to the revised problem?
- **Counter-Argument:** Does the team acknowledge the risks of their revised position and address them?
- **Decision Clarity:** Is the final recommendation unambiguous? Can the team state it in one sentence?

Submission Requirements

- Submit deliverables 1–5 as a single ZIP via the course LMS by the posted deadline (before class on presentation day).
- Name files: [TeamName]_ChatLog.pdf, [TeamName]_Storyboard.pdf, [TeamName]_Report.pdf, [TeamName]_Deck.pptx, [TeamName]_Reflection.pdf.
- One submission per team. Designate one person as the submitter.
- Late submissions: minus 10 points per day, no exceptions.
- Your Claude chat log must be from a single continuous conversation — do not stitch multiple conversations together.

- The Presentation Defense (Deliverable 6) occurs live in class. No additional submission required.

Grading Criteria

Total: 100 points

Component	Points	What We're Grading
Claude Process (Chat Log + Reflection)	**10**	All 15 prompts present; evidence of critical follow-up; reflection is specific and evaluative — not just a description of what was done
Storyboard (Minto / SCQA)	**10**	Clear SCQA structure; governing thought stated as a direct answer; supporting arguments are declarative and data-referenced
Written Executive Report	**30**	Level 2 and Level 3 thinking throughout; one governing thought; three data-grounded arguments; no unsupported claims; precise language
PowerPoint Presentation	**30**	Kawasaki format; one idea per slide; slide notes explain the talk track; narrative arc is clear and logically sequenced
Presentation Defense — Scenario Shock	**20**	Logic & Coherence; Evidence Application; Framework Usage; Counter-Argument acknowledgment; Decision Clarity — graded on reasoning quality, not polish

A Note on Academic Integrity and AI Use

Using Claude is required for this assignment. Using Claude to skip the analysis — pasting data and accepting a narrative without evaluating it — is not. The submitted chat log will make this distinction clear. More importantly, the Presentation Defense will make it unmistakable: students who did not genuinely engage with the data will not be able to answer questions about it in real time. The final decisions about what story to tell, how to tell it, and how to defend it under pressure are yours.

Appendix A: Claude Best Practices

- **Work with the raw file.** Don't pre-filter or summarize before uploading. Let Claude work with the full dataset.
- **Ask one thing at a time.** Compound questions produce muddier answers. Sequence your questions.
- **Use 'Challenge this.'** After Claude gives a finding, ask it to challenge its own output. This surfaces limitations you'd otherwise miss.

- **Specify what you want.** 'Tell me about the data' is worse than 'Show me the top 5 product lines by gross revenue with exact dollars and percentage of total.'
- **Don't accept the first draft.** Claude's first response is a starting point. Push it. Ask for more specificity, different framing, or a counterargument.
- **Paste your own work back in.** After writing a paragraph or argument, paste it back into Claude and ask it to evaluate it. This is one of its highest-value uses.
- **Prepare for the Shock.** Use Prompt 15 seriously. Think through what new constraints could destabilize your recommendation — and pre-stress-test your response.

Appendix B: Presentation Defense — What Good Looks Like

When the Shock is introduced, the difference between strong and weak teams is usually one of three things:

- **Strong teams:** State a clear revised position in the first 60 seconds and then support it with data.
- **Strong teams:** Acknowledge what the new constraint breaks in their original argument before explaining what still holds.
- **Strong teams:** Use specific product line names and revenue figures — not general language like 'the lower-performing products.'
- **Weak teams:** Spend the five minutes restating their original slides rather than responding to the new constraint.
- **Weak teams:** Say 'it depends' without specifying what it depends on or how they would find out.
- **Weak teams:** Refer to data vaguely — 'we saw some trends' — without being able to cite specific findings.

This assignment is designed so that Claude does what Claude is good at, and students do what students must learn to do. The habits you build here — interrogating data before narrativizing it, challenging AI outputs, defending a position when the ground shifts — are the habits that will distinguish you in the workforce.